

Regression Modelling and Classification

Group Assignment – Group 3

Note: Part A and Part B are two separate assignments (Assignment 3 – Regression, Assignment 4 – Classification). Each should be developed and submitted as its own repository – see Section 6.

1 Objective

An insurance company and a Human Resources department seek predictive models to improve pricing and employee retention decisions.

2 Dataset Description

Part A – Regression (Assignment 3)

Dataset: Medical Cost Personal Insurance – <https://www.kaggle.com/datasets/mirichoi0218/insurance>

Target Variable: Medical Insurance Charges

Predictors include:

- Age
- BMI
- Smoking Status
- Number of Children
- Region

Part B – Classification (Assignment 4)

Dataset: IBM HR Employee Attrition – <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Target Variable: Employee Attrition (Yes/No)

3 Analytical Tasks

Part A – Regression

1. Perform exploratory data analysis.
2. Build Simple and Multiple Linear Regression models.
3. Evaluate using: R^2 , Adjusted R^2 , RMSE, MAE.
4. Perform regression diagnostics.
5. Identify factors influencing insurance charges.
6. Recommend pricing strategies.

Part B – Classification

1. Prepare the dataset.
2. Build: Logistic Regression, Decision Tree.
3. Evaluate using: Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix.
4. Compare thresholds.
5. Identify important predictors of employee attrition.
6. Recommend HR intervention strategies.

4 Discussion

Answer the following:

1. Which factors most influence insurance costs?
2. Which regression model performs better?
3. Which classification model performs better?

4. Which threshold should HR adopt to identify at-risk employees?
5. How could AI improve the modelling process?

5 Deliverables

1. Python Notebook
2. Tables and Visualizations
3. Technical Report (4–6 pages)
4. Executive Presentation (8 slides maximum)

6 Repository Structure

Since Part A and Part B are separate assignments, maintain two independent repositories rather than one combined repo:

- **Assignment 3 – Regression:** notebook, EDA plots, regression diagnostics, report, and slides for the Medical Cost Personal Insurance dataset.
- **Assignment 4 – Classification:** notebook, EDA plots, model comparison, report, and slides for the IBM HR Employee Attrition dataset.

Each repo should have its own `README.md` (objective, dataset link, how to run), `requirements.txt`, and a `report/` and `slides/` folder, so the two assignments can be graded and versioned independently.